CropShield AI API Documentation

FastAPI route inventory and representative JSON contracts

Overview

FastAPI serves interactive OpenAPI documentation at **/docs** and ReDoc at **/redoc**. JSON request and response models are defined in app/models/schemas.py and individual route modules. Authentication uses bearer tokens for protected routes; the login response returns the token.

Primary application endpoints

Method and path	Purpose	Request / response summary
GET /health; GET /api/health	Service, retrieval and model readiness	JSON health state.
GET /api/crops; /api/states; /api/seasons	UI option metadata	JSON arrays of supported values.
POST /api/auth/login	Authenticate	{email, password} -> {success, token, user, message}.
POST /api/auth/signup	Create account	{name, email, password, role?} -> auth response; admin role cannot be self-assigned.
GET /api/auth/me	Validate current token	Authorization: Bearer -> {email, name, role}.
POST /api/query	Agricultural advisory	AdvisoryRequest -> recommendations, similar cases, risk/yield/impact, formatted advisory and latency.
POST /api/search	Hybrid case search	{query, crop?, state?, season?, ph?, rainfall?, temperature?, top_k?} -> list of cases.
GET /api/cases	Browse indexed cases	Optional crop/state/season/limit query parameters -> case list.
POST /api/recommend; /api/recommendation	Crop recommendation	Existing recommendation request schema -> RecommendationResponse.
POST /api/alternatives	Alternative crop recommendation	Existing recommendation request schema -> RecommendationResponse.
POST /api/predict-yield; /api/prediction	ML yield estimate	Yield prediction feature object -> predicted yield, interval and related fields.
POST /api/suitability	Crop suitability classification	Existing suitability inputs -> class and scores.
POST /api/analyze-impact	Agronomic impact analysis	Existing analysis input schema -> impact result.
POST /api/feedback; GET /api/feedback/summary	Feedback capture / aggregates	FeedbackRequest -> status; summary -> aggregate JSON.
POST /api/evaluate; GET /api/evaluation/latest	Evaluation run / saved result	Evaluation request -> report; latest -> saved JSON report.
POST /api/evaluation/crop; GET /api/evaluation/crop	Crop evaluation features	See generated OpenAPI schema for required fields.
POST /api/advisory-report/pdf	Generate advisory PDF	Existing advisory report payload -> downloadable PDF response.
/api/admin/*	Administrative overview, users, feedback, evaluation, ML, dataset and health	Admin routes; require authorized admin session/token.

Example requests

Advisory query:

```
{"query":"Compare rice and maize for Kharif in Andhra Pradesh with 1000 mm rainfall and pH 6.5","top_k":5}
```

Search cases:

```
{"query":"Wheat in Punjab during Rabi","crop":"Wheat","state":"Punjab","season":"Rabi","top_k":5}
```

Yield prediction:

```
{"crop":"Rice","season":"Kharif","state":"Andhra Pradesh","area":1000,"fertilizer":120,"pesticide":1,"n":80,"p":40,"k":40,"ph":6.5,"temperature":28,"rainfall":1100,"humidity":75}
```

Exact field constraints can evolve with the source schemas; run the server and consult **http://127.0.0.1:8000/docs** for the live contract. The endpoint list above reflects route decorators in app/api/.